

Power Functions & Function Comparison

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **identify** a power function and interpret the parameters in $y = kx^p$.

RELATED SKILLS

- I Interpret the coefficient and exponent in a power function.
- I Represent and interpret a power-variation relationship.

- ☐ 2. I can **construct** a power function from two data points or a proportional relationship.

RELATED SKILLS

- I Determine a power function from two points or a proportional relationship.
- A Isolate a specified variable in an equation or formula.

- ☐ 3. I can **relate** the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

RELATED SKILLS

- I Determine power-function behavior as the input approaches zero or positive or negative infinity.
- I Relate the coefficient and exponent of a power function to its graph shape.
- D Connect parameters in a formula to features of its graph.

- ☐ 4. I can **recognize and model** direct, inverse, and other power-variation relationships.

RELATED SKILLS

- I Represent and interpret a direct-variation relationship.
- I Represent and interpret an inverse-variation relationship.
- D Represent and interpret a power-variation relationship.

- ☐ 5. I can **solve** equations involving integer or rational powers.

RELATED SKILLS

- A Interpret numerator and denominator roles in a rational exponent.
- A Use inverse operations to maintain an equivalent equation.

- ☐ 6. I can **compare** the long-run dominance of logarithmic, power, polynomial, and exponential functions.

RELATED SKILLS

- I Compare the long-run growth or decay of functions from different families.
- D Compare functions shown in different representations.

☐ **7. I can **fit** a power regression model and interpret its parameters and correlation evidence.**

RELATED SKILLS

- I** Use technology to determine a power regression model.
 - D** Interpret the coefficient and exponent in a power function.
 - A** Interpret the sign and magnitude of a correlation coefficient in context.
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